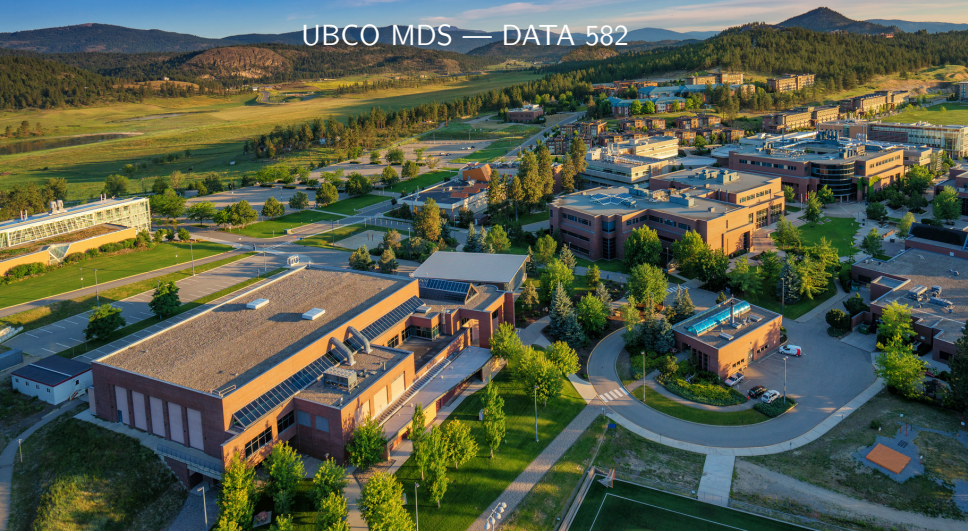


Problems and Solutions

UBCO MDS — DATA 582



- ▶ Recall the simplest form for us to consider:

$$h(\theta \mid \mathbf{x}) \propto f(\mathbf{x} \mid \theta)g(\theta)$$

- ▶ The **posterior** distribution of θ depends on the observed data (through the **likelihood**) and the **prior** distribution we imposed on θ .
- ▶ We have covered the Beta-Bernoulli(Binomial) and Normal-Normal cases so far. These were conveniently special cases of **conjugate priors**.
- ▶ Let's discuss more arbitrary prior-and-likelihood choices, and some concerns that then arise.

- ▶ What if we assume
 - ▶ $X \sim N(\mu, \sigma^2)$ with σ^2 known.
 - ▶ $\mu \sim \text{Beta}(\alpha, \beta)$
- ▶ This suggests that we **know** that μ is bounded $[0,1]$, and we could set α and β to assume more regarding the shape of the distribution within those bounds.
- ▶ Perhaps a little wonky of an assumption, though keep in mind that we could add some more parameters to generalize the Beta to different bounds.

► Reminder: $h(\mu \mid \mathbf{x}) \propto f(\mathbf{x} \mid \mu)g(\mu)$

► So we multiply:

$$\begin{aligned} h(\theta \mid \mathbf{x}) &\propto \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (x_i - \mu)^2 \right\} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \mu^{\alpha-1} (1 - \mu)^{\beta-1} \\ &\propto \exp \left\{ -\frac{1}{2} \frac{(\mu - \bar{x})^2}{\sigma^2/n} \right\} \mu^{\alpha-1} (1 - \mu)^{\beta-1} \end{aligned}$$

► Functional form of μ ?

No Functional Form?



- ▶ If the functional form is not recognizable, it is not immediately clear what the posterior distribution is
- ▶ Ultimately, we're only missing a constant out front that ensures $h(\mu \mid \mathbf{x})$ is a true probability distribution
- ▶ But without that, we would struggle to do most inferential steps...for example, the point estimate using the posterior mean:

$$E[\mu] = \int \mu h(\mu \mid \mathbf{x}) d\mu$$

Finding the Constant?



- ▶ We could try to compute the constant:

$$\int_{\Omega} f(\mathbf{x} \mid \mu) g(\mu) d\mu$$

- ▶ But integration can be hard (and even impossible explicitly)...
- ▶ We will need a different approach...that you've already been exposed to (580/581)



- ▶ Let's back off Bayesian momentarily, and simplify our issue.
- ▶ Pretend we have a known probability distribution (it integrates to 1 across the domain) that is tricky to solve definite integrals for.
- ▶ It shouldn't be too hard to imagine. The normal distribution is not terribly simple to integrate! Feel free to look up error functions and approximations thereof.
- ▶ There's a reason we provide normal probability tables in intro stats.



- ▶ But suppose we were naive to any integration tricks, but were able to simulate observations from the target distribution.
- ▶ Example: we know $X \sim A(\theta)$ and want to find $P(X < 0)$, but don't know this 'A' distribution well (is it skewed? symmetric? etc), cannot directly integrate it, and there's no known literature/methods. If we can generate observations from $A(\theta)$, what should we do?
- ▶ Recall: This general method is called **Monte Carlo approximation/integration**.
- ▶ We can approximate properties of a continuous-valued function using discrete-valued observations. Allowing us to move from an integral (hard) to a sum (easy).

Recall: Monte Carlo



- ▶ More generally, suppose $X \sim A(\theta)$ suggests the pdf $f(x | \theta)$. We are interested in $E[d(x)]$, which is the expected value of some function d of x . Then we need

$$E[d(x)] = \int d(x)f(x | \theta)dx$$

- ▶ We cannot do the integral, but we can sample n values $x_1, \dots, x_n$ so we propose approximating via

$$\bar{d}_n = \frac{\sum_{i=1}^n d(x_i)}{n}$$

- ▶ If x_i are independent samples, then the LLN states

$$\bar{d}_n \rightarrow E[d(x)] \quad \text{as} \quad n \rightarrow \infty$$

- ▶ In other words, larger sample means better approximation.



- ▶ Mixture models are a weighted sum of probability distributions. Suppose

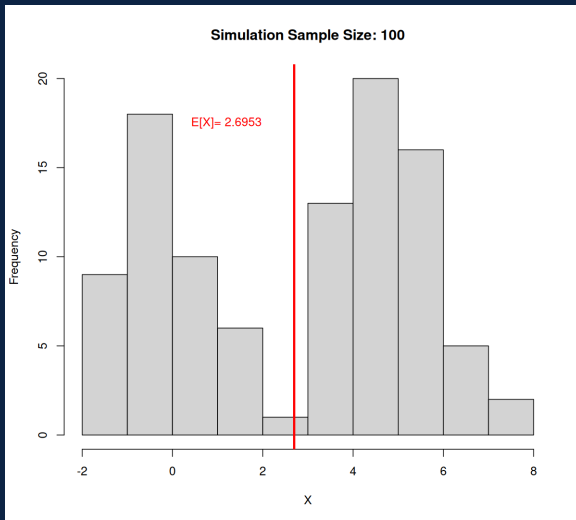
$$f(x) = 0.4\phi(x \mid \mu_1 = 0, \sigma_1^2 = 1) + 0.6\phi(x \mid \mu_2 = 5, \sigma_2^2 = 1)$$

- ▶ Simulating observations from this distribution is easy
- ▶ We could approximate $E[X] \approx \frac{\sum x_i}{n}$ using Monte Carlo integration...

Mixtures - Expected Value



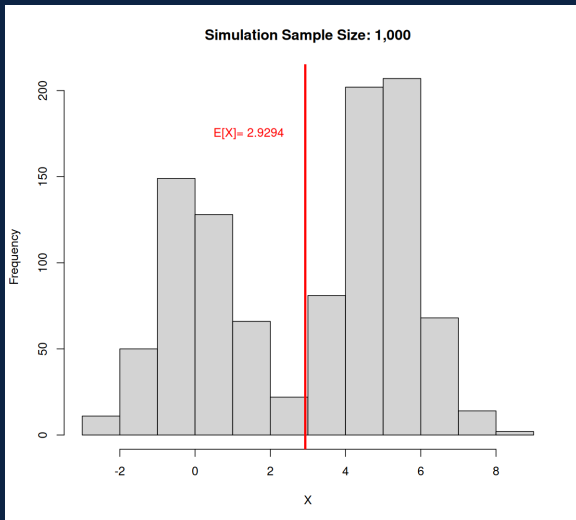
- Note via conditional expectations $E[X] = 0.4(0) + 0.6(5) = 3$



Mixtures - Expected Value



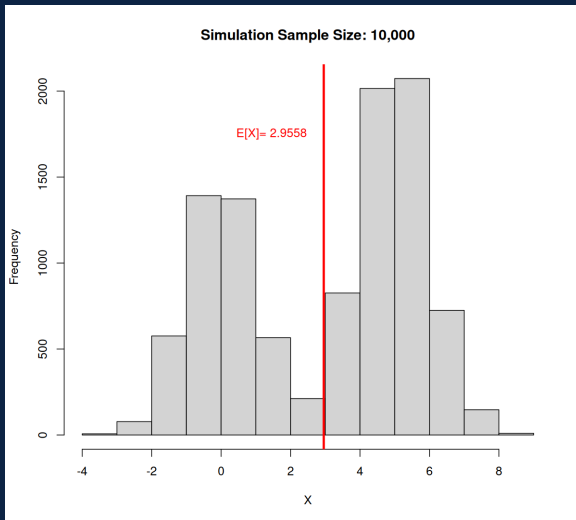
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Mixtures - Expected Value



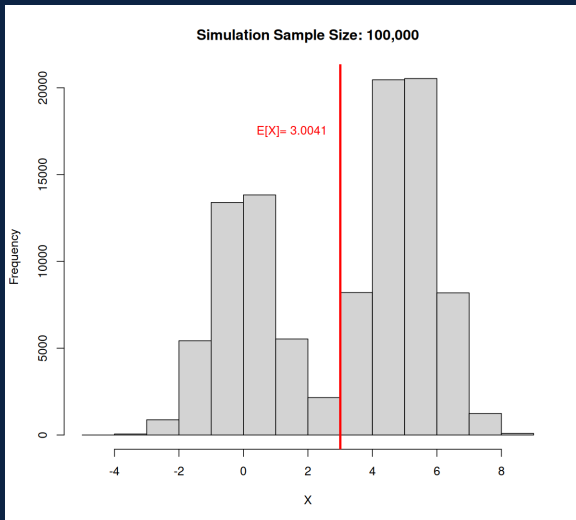
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Mixtures - Expected Value



- Note via conditional expectations $E[X] = 0.4(0) + 0.6(5) = 3$





- ▶ Great, so if we have a known probability distribution, MC approximation can help us perform complex inferences without any need to integrate.
- ▶ R has many built in functions for generating from common distributions. Often though, even if we have sorted out the probability distribution of interest, independent sampling from the distribution can be challenging.
- ▶ Also, we haven't yet addressed the fact that we only know our $h(\theta \mid \mathbf{x})$ proportionally!

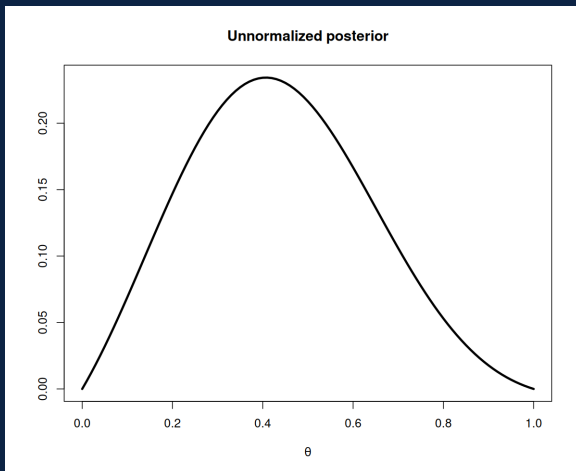
- ▶ **Rejection sampling** is a generalized technique for acquiring samples from a target distribution by using a proposal distribution.
- ▶ With some unnormalized target density $h_{\infty}(\theta \mid \mathbf{x})$ to sample from, we need some constant M and (easy to sample from) proposal density $w(\theta)$ such that $h_{\infty}(\theta \mid \mathbf{x}) \leq Mw(\theta)$ for all θ
- ▶ We then
 1. Sample θ from $w(\theta)$
 2. Sample U from $\text{Uniform}(0,1)$
 3. Compute the ratio $R = \frac{h_{\infty}(\theta \mid \mathbf{x})}{w(\theta)}$
 4. If $R < MU$ then reject θ , else accept and include in your sample.

- ▶ Reminder: $h(\mu \mid \mathbf{x}) \propto f(\mathbf{x} \mid \mu)g(\mu)$
- ▶ We found

$$h(\theta \mid \mathbf{x}) \propto \exp \left\{ -\frac{1}{2} \frac{(\mu - \bar{x})^2}{\sigma^2/n} \right\} \mu^{\alpha-1} (1 - \mu)^{\beta-1}$$

- ▶ Suppose we use $\alpha = 2$, $\beta = 2$ in our prior and observe $\bar{x} = 0.33$ from $n = 10$ samples...

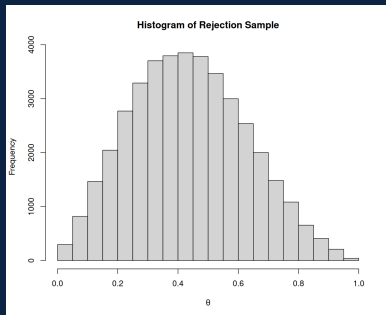
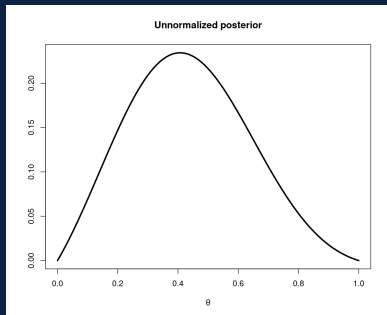
- We can plot this unnormalized posterior





- ▶ So, we can set up our rejection sample with, say $M=0.3$
 1. Sample θ from $\text{Uniform}(0,1)$
 2. Sample U from $\text{Uniform}(0,1)$
 3. Compute the ratio $R = \frac{h_\infty(\theta|\mathbf{x})}{1}$
 4. If $R < 0.3U$ then reject θ , else accept and include in your sample.

- ▶ Giving us the following rejection sample from 100,000 attempts (accepted sample size 40,696)





- ▶ Rejection sampling (and other general techniques for acquiring iid samples) can be extremely inefficient for many target distributions
- ▶ Reason being that your rejection rate climbs if, say, your target distribution has sharp peaks.
- ▶ Enter Markov chain-based methods for Monte Carlo approximation! We won't review a bunch of details here from 581.



- ▶ The Metropolis-Hastings algorithm is amongst the more important statistical developments of the last 60ish years.
- ▶ It provides a general way to construct MCMC chains that provide our target distribution
- ▶ Recall: Markov chains are a sequence of dependent random variables (aka, not independent ones). However, if the chain reaches a stationary/equilibrium/convergence, then the sample across this path can be used for the same Monte Carlo purposes.

- ▶ The Metropolis-Hastings algorithm generates a chain via
 1. Given a starting position θ^t , generate a candidate value y via a proposal density $w(y | \theta^t)$
 2. Calculate acceptance probability

$$a(y | \theta^t) = \min \left(1, \frac{h_{\infty}(y | \mathbf{x}) w(\theta^t | y)}{h_{\infty}(\theta^t | \mathbf{x}) w(y | \theta^t)} \right)$$

3. With probability a , set $\theta^{t+1} = y$, else set $\theta^{t+1} = \theta^t$
 4. Return to 1 until a set number of iterations are run.
- ▶ Choice of w is important here, it has to satisfy the potential domain of θ but if poorly chosen the number of rejections can be high.
 - ▶ If w is symmetric ($w(\theta^t | y) = w(y | \theta^t)$) then the acceptance probability simplifies to only depend on h_{∞}



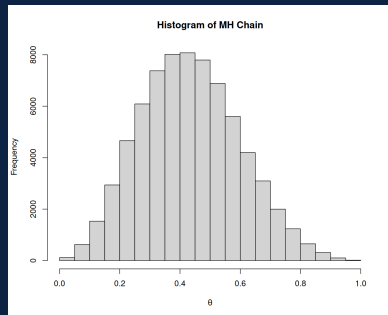
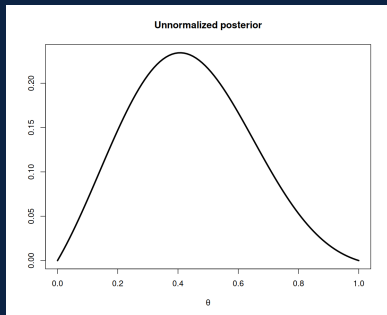
► We set up our chain as...

1. Given a starting position $\theta^t = 0.2$, generate a candidate value y via a proposal density $w(y | \theta^t) = \theta^t + Z$ where $Z \sim N(0, 0.2^2)$
2. Calculate acceptance probability

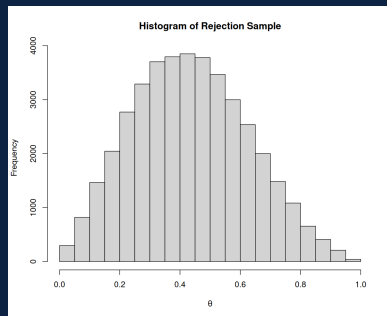
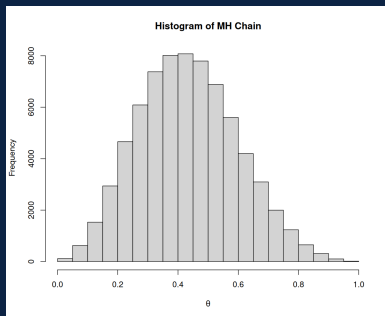
$$a(y | \theta^t) = \min \left(1, \frac{h_\infty(y | \mathbf{x})}{h_\infty(\theta^t | \mathbf{x})} \right)$$

3. With probability a , set $\theta^{t+1} = y$, else set $x^{t+1} = x^t$
4. Return to 1 until a set number of iterations are run.

- ▶ Giving us the following MCMC sample from 100,000 attempts (accepted sample size 71,319)



- ▶ With either of these MC samples, we can carry out approximate inference. Credible intervals using quantiles, for example.





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